

The Wind-Powered Sail Car

Big challenge · Outdoors · Grown-up hands-on · Ages 7–11 · 45 min

Brought to you by Chike — “Hmmm, what should we create today?”

Build a car with no engine, then find out what shape of sail carries it furthest.

WHAT YOU NEED

- Cardboard for the body
- 4 bottle caps or card circles for wheels
- 2 skewers or straws for axles
- Paper for sails
- Tape
- A hand fan or a breezy day

BEFORE YOU START

A grown-up handles the skewers and caps the ends.

Test in an open space away from roads.

How to make it

- 1 Cut a rectangle of cardboard for the car body.
- 2 Tape two straws underneath as axle housings, then thread the skewers through and fix a wheel on each end. Check they spin freely.
- 3 Tape a mast upright in the middle of the body.
- 4 Cut a square sail, tape it to the mast, and test it. Mark where the car stopped.
- 5 Swap the square for a triangle of the same size and test again from the same start line. Mark it.
- 6 Try a third shape of your own. Measure all three distances and decide which sail won.

TALK ABOUT IT

Your three sails were the same size but traveled different distances. What else, apart from size, seems to matter?

TRY THIS TOO

Keep the winning sail and change the wheels instead. Wide wheels, narrow wheels, four wheels, three.

DID YOU KNOW? · WEATHER & AIR

Wind pushes a sail, and the sail pushes the car. Moving air is made of countless tiny particles bumping into whatever is in the way. A sail gives them a big surface to push against, and that push travels through the mast into the car.